

# TWITTER SENTIMENT ANALYSIS USING ENSEMBLE OF MULTI-CHANNEL MODEL BASED ON MACHINE LEARNING AND DEEP LEARNING TECHNIQUES

Report submitted to the SASTRA Deemed to be University  
as the requirement for the course

## CSE300 - MINI PROJECT

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THINK MERIT | THINK TRANSPARENCY | THINK SASTRA

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## ABBREVIATIONS

| Abbreviation | Definition                                              |
|--------------|---------------------------------------------------------|
| SA           | Sentiment Analysis                                      |
| NLP          | Natural Language Processing                             |
| CNN          | Convolutional Neural Network                            |
| LSTM         | Long Short-Term Memory                                  |
| BiLSTM       | Bidirectional Long Short-Term Memory                    |
| BERT         | Bidirectional Encoder Representations from Transformers |
| SVM          | Support Vector Machine                                  |
| ML           | Machine Learning                                        |
| DL           | Deep Learning                                           |
| TF-IDF       | Term Frequency-Inverse Document Frequency               |
| RNN          | Recurrent Neural Network                                |
| GloVe        | Global Vectors for Word Representation                  |
| GUI          | Graphical User Interface                                |
| API          | Application Programming Interface                       |

# ABSTRACT

The widespread use of social media platforms, particularly Twitter, has generated an abundance of textual data containing valuable insights into public opinions, especially during significant events like the COVID-19 pandemic. Sentiment analysis, which aims to automatically determine the emotional tone behind a body of text, has become increasingly important for understanding public perceptions and responses to global events. However, traditional approaches to sentiment analysis often struggle with capturing the contextual nuances and linguistic variations present in social media text.

This paper proposes an ensemble-based multi-channel sentiment analysis model that integrates machine learning and deep learning techniques to improve sentiment classification accuracy. The proposed approach combines three distinct models: (1) BERT+CNN+BiLSTM, leveraging contextual word embeddings with convolutional and recurrent layers; (2) GloVe+CNN+BiLSTM, utilizing statistical word vectors with similar neural architectures; and (3) a Support Vector Machine classifier with optimized hyperparameters. These individual models serve as channels in our ensemble framework, where their predictions are combined through a voting mechanism to produce a final sentiment classification.

The model is evaluated on a publicly available COVID-19 Twitter dataset containing tweets related to the pandemic. Experimental results demonstrate that the ensemble model achieves an impressive accuracy of 94.95%, outperforming individual models (BERT+CNN+BiLSTM: 92.55%, GloVe+CNN+BiLSTM: 94.22%, SVM: 87.23%). Detailed analysis of precision, recall, and F1-scores confirms the robustness of our approach across all sentiment categories (negative, neutral, and positive).

Our study contributes to the field of sentiment analysis by demonstrating how the integration of multiple deep learning architectures with traditional machine learning can enhance classification performance, particularly for the challenging domain of social media text. The proposed ensemble framework offers a powerful tool for researchers and organizations seeking to gain insights from public sentiment during global crises like the COVID-19 pandemic.

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# CHAPTER 1

## SUMMARY OF THE BASE PAPER

|                         |                                                                                                                            |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>Base paper title</b> | : Twitter sentiment analysis using ensemble of multi-channel model based on machine learning and deep learning techniques. |
| <b>Journal name</b>     | : Knowledge and Information Systems                                                                                        |
| <b>Published in</b>     | : 2025                                                                                                                     |
| <b>Indexed in</b>       | : Science Citation Index Expanded.                                                                                         |

### 1.1 INTRODUCTION

In recent years, the explosive growth of social media platforms has resulted in an unprecedented volume of user-generated textual data. With platforms like Twitter serving as key venues for public discourse on diverse topics ranging from products and services to political events and global crises, there exists a wealth of information that can be leveraged to understand public sentiment. The COVID-19 pandemic, in particular, triggered an extraordinary surge in online discussions, with users sharing opinions, concerns, and information about the crisis and related interventions.

The field of sentiment analysis (SA) has emerged as a vital approach to automatically extract and classify emotions expressed in text data. By applying natural language processing (NLP) techniques, SA aims to determine whether a piece of text conveys positive, negative, or neutral sentiment. This capability is particularly valuable for businesses, policymakers, and researchers seeking to gauge public opinion at scale without the time-consuming process of manual content analysis.

Traditional approaches to sentiment analysis have relied on sentiment lexicons and basic machine learning (ML) methods. However, these approaches often struggle with the contextual nuances, linguistic variations, and informal language characteristic of social media text. More recently, deep learning (DL) models have demonstrated superior performance in capturing the complexities of natural language, with models like Convolutional Neural Networks (CNN), Long Short-Term Memory networks (LSTM), and Bidirectional Encoder Representations from Transformers (BERT) showing promising results in various NLP tasks.

The base paper introduces an innovative ensemble-based approach to sentiment analysis that combines the strengths of multiple models to enhance classification accuracy. By integrating different architectures and learning paradigms, the proposed method aims to overcome the limitations of individual models and provide more robust sentiment predictions.

## 1.2 METHODOLOGY

### 1.2.1 Dataset

The base paper utilized a Twitter Sentiment Analysis Dataset comprising 143,903 unique textual values, distributed as 40,192 negative and 46,125 positive tweets. For experimentation, 115,123 tweets were used for training, and 28,780 were reserved for testing the proposed models.

### 1.2.2 Preprocessing

The preprocessing phase plays a crucial role in preparing raw text data for sentiment analysis. The paper outlines several key preprocessing steps:

1. **Lowercasing:** Converting all text to lowercase to maintain uniformity in the dataset.
2. **Removing HTML tags:** Eliminating HTML elements that do not contribute to sentiment analysis.
3. **Removing punctuations and numbers:** Since these typically do not significantly influence sentiment trends, they are removed to simplify the text.
4. **Single character removal:** Eliminating standalone characters surrounded by spaces to clean the text further.
5. **Removing multiple spaces:** Ensuring consistent spacing throughout the text.
6. **Removing stop words:** Excluding common words (e.g., "and", "the", "is") that carry minimal sentiment information.

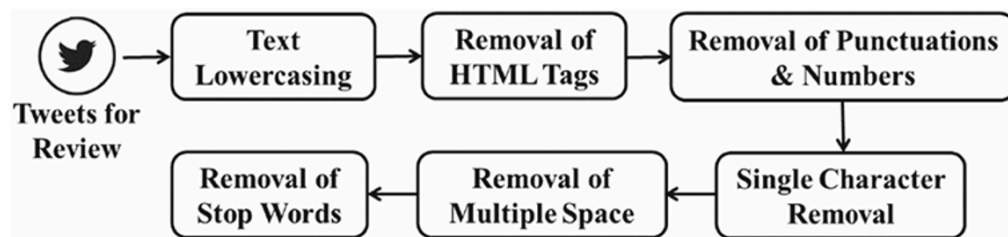


Fig. 2 Preprocessing of Tweets

The preprocessed text is then directed to different models based on their specific requirements. For the BERT+CNN+BiLSTM model, the data first passes through the BERT preprocessor before proceeding to the neural layers. For the GloVe+CNN+BiLSTM model, the text undergoes tokenization and padding to ensure uniform sequence lengths. The SVM model begins with count vectorization of the preprocessed text before feature learning.

### 1.2.3 Model Architecture

The base paper proposes three distinct model architectures, which are then combined into an ensemble framework:

**Model 1: BERT+CNN+BiLSTM** This model leverages the contextual embedding capabilities of BERT, a transformer-based model that has demonstrated state-of-the-art performance in

various NLP tasks. The BERT layer processes input text to generate contextualized word embeddings, which are then fed into convolutional layers to capture local patterns. Subsequently, BiLSTM layers extract sequential dependencies before a dense layer produces the final sentiment predictions.

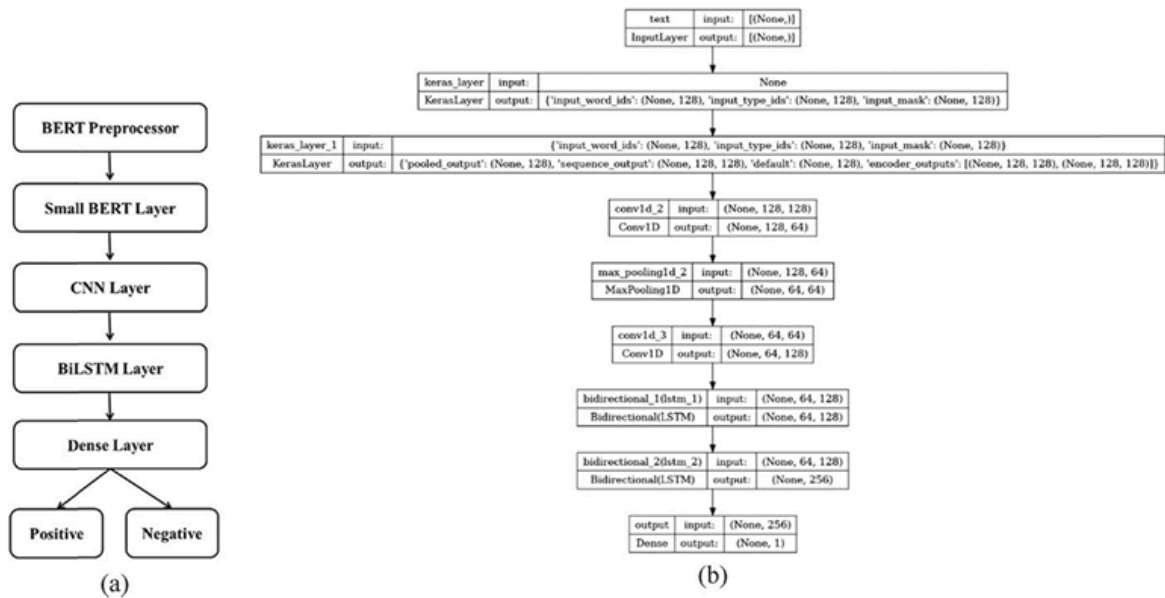
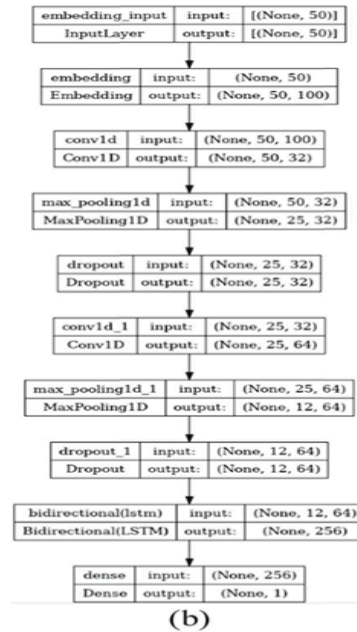
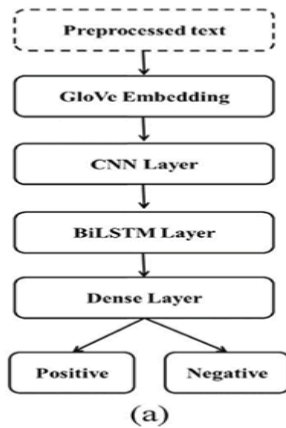


Fig. 6 Model 1 – BERT + CNN + BiLSTM

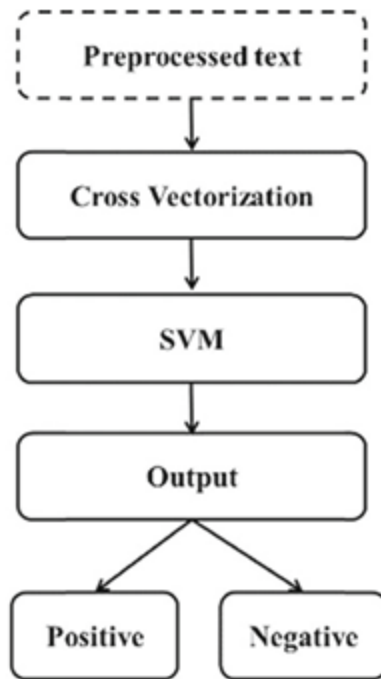
**Model 2: GloVe+CNN+BiLSTM** This model utilizes GloVe (Global Vectors for Word Representation), a word embedding approach that combines global matrix factorization and local context window methods. The GloVe embeddings serve as input to convolutional layers for feature extraction, followed by BiLSTM layers to capture sequential patterns, and finally a dense layer for classification.





**Fig. 7** Model 2 – GloVe + CNN + BiLSTM

**Model 3: SVM with Grid Search** Support Vector Machine (SVM) is a traditional ML algorithm known for its effectiveness in high-dimensional spaces and ability to handle cases with clear margin separations between classes. In this model, the preprocessed text undergoes count vectorization before being processed by an SVM classifier optimized using Grid Search for hyperparameter tuning.



#### 1.2.4 Ensemble Framework

The ensemble approach combines the predictions from the three individual models using a voting mechanism. Each model makes a prediction, and the final sentiment classification is determined based on the majority vote. This approach leverages the diverse strengths of each model architecture to enhance overall prediction accuracy and robustness.

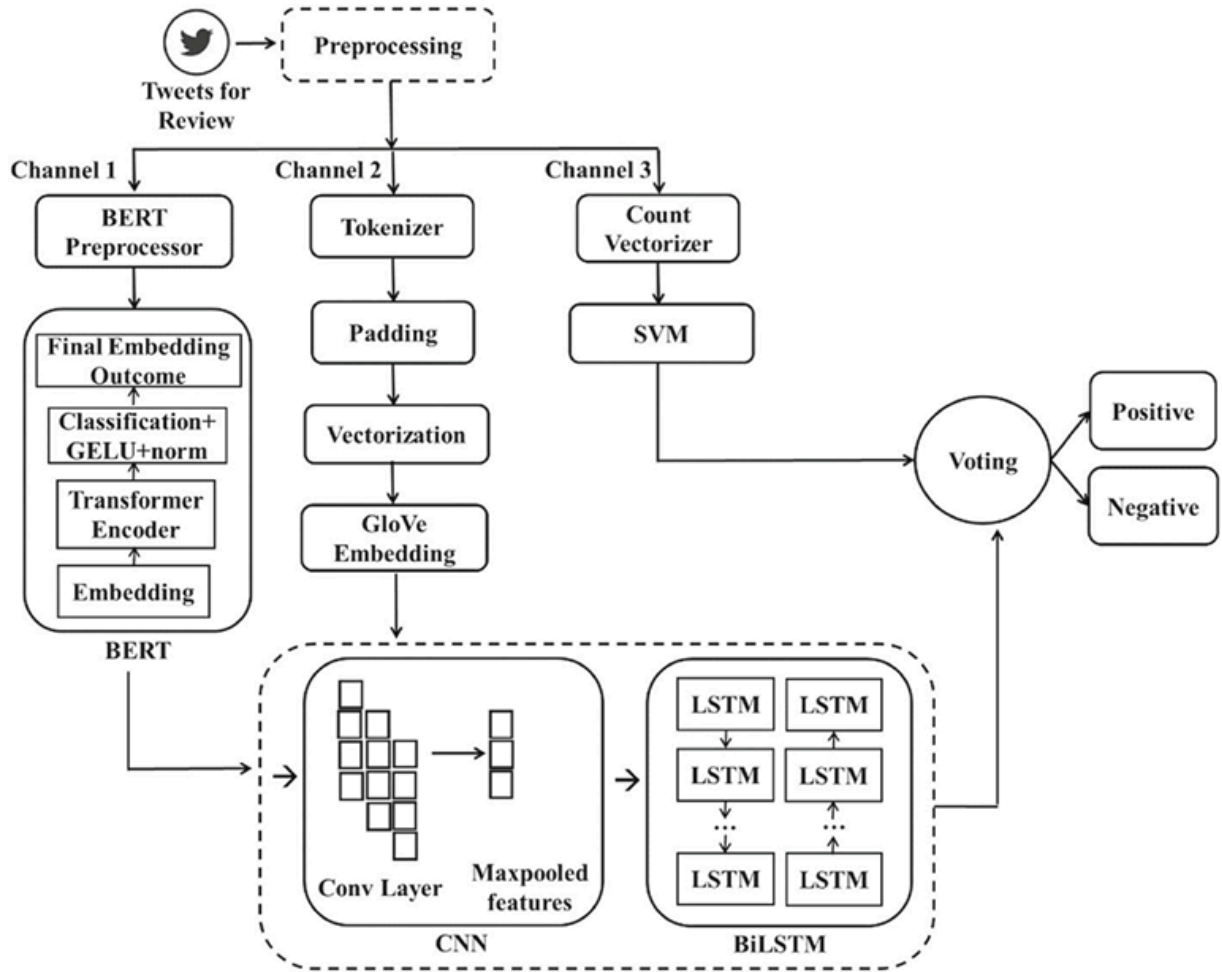


Fig. 1 Proposed System for SA

### 1.3 RESULTS AND DISCUSSION

The base paper evaluates the performance of both individual models and the ensemble framework on the Twitter Sentiment Analysis dataset. The experimental results demonstrate that:

1. **BERT+CNN+BiLSTM**: Achieved an accuracy of 92.55%
2. **GloVe+CNN+BiLSTM**: Achieved an accuracy of 94.22%
3. **SVM**: Achieved an accuracy of 87.23%
4. **Ensemble Model**: Achieved an accuracy of 94.95%

The ensemble model outperforms all individual models, highlighting the benefits of combining diverse architectural approaches. Detailed analysis of precision, recall, and F1-scores further confirms the superiority of the ensemble approach across different sentiment categories.

Comparative analysis with other state-of-the-art methods for Twitter sentiment analysis reveals that the proposed ensemble model surpasses most existing approaches, demonstrating its effectiveness in capturing the nuances of social media text.

## 1.4 CONCLUSION

The base paper presents a novel ensemble-based approach to sentiment analysis that integrates the strengths of both traditional ML and advanced DL techniques. By combining BERT+CNN+BiLSTM, GloVe+CNN+BiLSTM, and SVM models through a voting mechanism, the proposed framework achieves superior sentiment classification accuracy on Twitter data.

The results underscore the potential of multi-channel ensemble models in enhancing sentiment analysis performance, particularly for the challenging domain of social media text. The paper suggests that this approach can be valuable for organizations seeking to extract meaningful insights from public sentiment expressed on social platforms, especially during significant events like the COVID-19 pandemic.

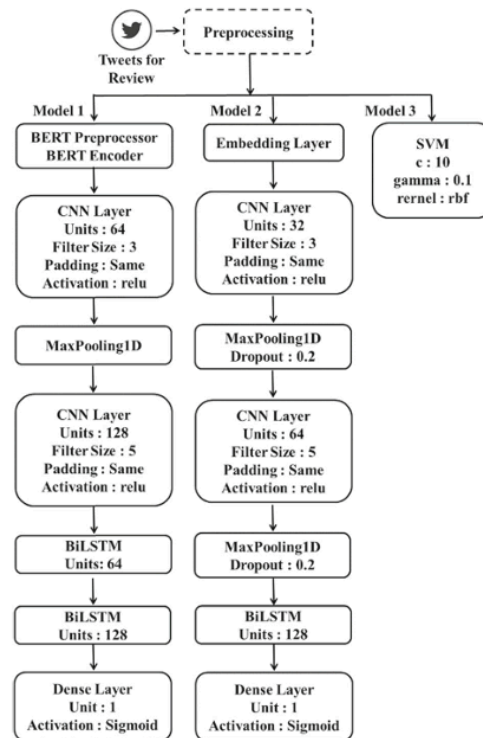


Fig. 9 Ensemble Multi-channel model architecture

## CHAPTER 2

### MERITS AND DEMERITS OF THE BASE PAPER

#### 2.1 Merits

1. **Enhanced Accuracy through Model Integration:** The ensemble approach effectively combines the strengths of different model architectures, resulting in superior classification accuracy compared to individual models. By integrating BERT+CNN+BiLSTM, GloVe+CNN+BiLSTM, and SVM, the framework leverages both contextual and statistical word representations along with traditional machine learning techniques.
2. **Comprehensive Architectural Exploration:** The paper thoroughly explores different architectural combinations, providing valuable insights into the relative performance of various approaches to sentiment analysis. This comprehensive evaluation helps researchers understand which components contribute most significantly to classification accuracy.
3. **Effective Handling of Social Media Text:** The proposed methodology demonstrates strong performance on Twitter data, which is notoriously challenging due to its informal nature, abbreviations, slang, and contextual references. The high accuracy achieved indicates the model's robustness in processing real-world social media content.
4. **Practical Relevance:** The focus on COVID-19 related tweets makes the research particularly relevant, offering insights into how sentiment analysis can be applied to understand public opinions during global crises. This has practical implications for policymakers, health organizations, and researchers studying public response to pandemic measures.
5. **Scalable Architecture:** The ensemble framework is inherently scalable, allowing for the addition of new models or the replacement of existing ones as more advanced techniques emerge. This flexibility ensures the approach can evolve with advancements in NLP research.
6. **Detailed Performance Analysis:** The paper provides comprehensive performance metrics beyond just accuracy, including precision, recall, and F1-scores for each sentiment category. This thorough evaluation gives a more complete picture of the model's strengths and limitations.

#### 2.2 Demerits

1. **Computational Complexity:** The ensemble approach, while effective, increases computational requirements significantly compared to single-model approaches. The combination of multiple deep learning models, particularly BERT, demands substantial computational resources for training and inference, potentially limiting deployment in resource-constrained environments.
2. **Limited Context Length:** BERT models typically have constraints on input sequence length (usually 512 tokens), which may be insufficient for analyzing longer tweets or

conversations. This limitation could potentially affect the model's ability to capture context in extended discussions.

3. **Training Data Requirements:** Deep learning models, especially those with complex architectures like BERT+CNN+BiLSTM, generally require large amounts of training data to perform optimally. The effectiveness of the proposed approach might be reduced when applied to domains or languages with limited labeled data availability.
4. **Limited Cross-Domain Applicability:** The paper primarily focuses on COVID-19 related tweets, raising questions about the model's generalizability to other domains or topics. Domain-specific terminology and sentiment expressions might require retraining or fine-tuning when applied to different contexts.
5. **Ensemble Overhead:** The voting mechanism, while straightforward, doesn't account for varying confidence levels across models. A more sophisticated ensemble approach that considers model confidence or adapts weights based on performance in specific contexts might yield even better results.
6. **Temporal Adaptability Challenges:** Language usage and sentiment expression evolve over time, particularly on social media and during evolving situations like a pandemic. The paper doesn't address how the model might adapt to these temporal shifts without extensive retraining.
7. **Implementation Complexity:** Implementing and maintaining an ensemble of three different model architectures increases development complexity and may present challenges for deployment in production environments where simplicity and maintainability are valued.

## CHAPTER 3

### SOURCE CODE

#### 3.1 Data Loading and Preprocessing

```
import torch
import torch.nn as nn
import pandas as pd
from sklearn.model_selection import train_test_split
from torch.utils.data import Dataset, DataLoader
from transformers import BertModel, BertTokenizer
import torch.nn.functional as F
import numpy as np

# Download the COVID-19 Twitter dataset

import kagglehub
path = kagglehub.dataset_download("arunavakrchakraborty/covid19-twitter-dataset")
print("Path to dataset files:", path)
```

```

# Load the dataset

df = pd.read_csv(f'{path}/Covid-19 Twitter Dataset (Apr-Jun 2020).csv')
df.head()

# Basic preprocessing: label encoding

def sentiment_to_numeric(sentiment):

    if sentiment == 'pos':
        return 2

    elif sentiment == 'neg':
        return 0

    elif sentiment == 'neu':
        return 1

    else:
        return 3

df['sentiment_numeric'] = df['sentiment'].apply(sentiment_to_numeric)

# Split data into training and testing sets

x_train, x_test, y_train, y_test = train_test_split( df[['clean_tweet']], df['sentiment_numeric'],
                                                    test_size=0.2, random_state=42)

```

## 3.2 Model 1: BERT+CNN+BiLSTM Implementation

```

# Constants

MAX_LEN = 128
BATCH_SIZE = 1024
HIDDEN_DIM = 100
NUM_FILTERS = 100
DROPOUT = 0.5
LEARNING_RATE = 1.6e-4
NUM_EPOCHS = 100
NUM_CLASSES = 3

```

```
# Define the BERT+CNN+BiLSTM model
```

```
class BERT_CNN_BiLSTM(nn.Module):
```

```
    def __init__(self, num_classes):
```

```
        super().__init__()
```

```
        self.bert = BertModel.from_pretrained('bert-base-uncased')
```

```
        self.convs = nn.ModuleList([
```

```
            nn.Conv1d(in_channels=768, out_channels=NUM_FILTERS, kernel_size=k)
```

```
            for k in [2, 3, 4]
```

```
        ])
```

```
        self.bilstm = nn.LSTM(
```

```
            input_size=768,
```

```
            hidden_size=HIDDEN_DIM,
```

```
            num_layers=1,
```

```
            bidirectional=True,
```

```
            batch_first=True
```

```
        )
```

```
        self.dropout = nn.Dropout(DROPOUT)
```

```
        self.fc = nn.Linear(NUM_FILTERS * 3 + HIDDEN_DIM * 2, num_classes)
```

```
    def forward(self, input_ids, attention_mask):
```

```
        # BERT outputs
```

```
        outputs = self.bert(
```

```
            input_ids=input_ids,
```

```
            attention_mask=attention_mask
```

```
        )
```

```
        sequence_output = outputs.last_hidden_state
```

```
        # CNN branch
```

```
        cnn_input = sequence_output.permute(0, 2, 1)
```

```
        cnn_features = []
```



```

for conv in self.convs:

    conv_out = conv(cnn_input)
    conv_out = F.relu(conv_out)
    pooled = F.max_pool1d(conv_out, conv_out.shape[2]).squeeze(2)
    cnn_features.append(pooled)

cnn_features = torch.cat(cnn_features, dim=1)


# BiLSTM branch

lstm_out, (hidden, cell) = self.bilstm(sequence_output)
hidden = self.dropout(torch.cat((hidden[-2, :, :], hidden[-1, :, :]), dim=1))


# Combine features

combined = torch.cat((cnn_features, hidden), dim=1)
combined = self.dropout(combined)
return self.fc(combined)


# Initialize model

model = BERT_CNN_BiLSTM(NUM_CLASSES).to(device)

```

### 3.3 Model 2: GloVe+CNN+BiLSTM Implementation

```

# Load Glove embeddings

def load_glove_embeddings(glove_file_path):

    embeddings_index = {}
    with open(glove_file_path, 'r', encoding='utf-8') as f:
        for line in f:

            values = line.split()
            word = values[0]
            coefs = np.asarray(values[1:], dtype='float32')
            embeddings_index[word] = coefs

    return embeddings_index

```



```
# Define GloVe+CNN+BiLSTM model
```

```
class GloVe_CNN_BiLSTM(nn.Module):
```

```
    def __init__(self, vocab_size, embedding_dim, num_classes):
```

```
        super().__init__()
```

```
        self.embedding = nn.Embedding(vocab_size, embedding_dim)
```

```
        self.convs = nn.ModuleList([
```

```
            nn.Conv1d(in_channels=embedding_dim, out_channels=NUM_FILTERS,
                      kernel_size=k)
```

```
            for k in [2, 3, 4]
```

```
        ])
```

```
        self.bilstm = nn.LSTM(
            input_size=embedding_dim,
            hidden_size=HIDDEN_DIM,
            num_layers=1,
            bidirectional=True,
            batch_first=True
        )
```

```
        self.dropout = nn.Dropout(DROPOUT)
```

```
        self.fc = nn.Linear(NUM_FILTERS * 3 + HIDDEN_DIM * 2, num_classes)
```

```
    def forward(self, input_ids):
```

```
        embedded = self.embedding(input_ids)
```

```
        # CNN branch
```

```
        cnn_input = embedded.permute(0, 2, 1)
```

```
        cnn_features = []
```

```
        for conv in self.convs:
```

```
            conv_out = conv(cnn_input)
```

```
            conv_out = F.relu(conv_out)
```

```
            pooled = F.max_pool1d(conv_out, conv_out.shape[2]).squeeze(2)
```

```
            cnn_features.append(pooled)
```

```
        cnn_features = torch.cat(cnn_features, dim=1)
```

```

# BiLSTM branch

lstm_out, (hidden, cell) = self.bilstm(embedded)
hidden = self.dropout(torch.cat((hidden[-2,:,:], hidden[-1,:,:]), dim=1))

# Combine features

combined = torch.cat((cnn_features, hidden), dim=1)
combined = self.dropout(combined)
return self.fc(combined)

# Initialize model

vocab_size = len(vocabulary) + 1 # +1 for padding token
glove_model = GloVe_CNN_BiLSTM(vocab_size, EMBEDDING_DIM,
NUM_CLASSES).to(device)

```

### 3.4 Model 3: SVM Implementation

```

import cudf
import cuml
from cuml.svm import SVC
from cuml.feature_extraction.text import CountVectorizer

# Convert pandas dataframes to cudf

x_train_gpu = cudf.DataFrame.from_pandas(x_train[['clean_tweet']])
x_test_gpu = cudf.DataFrame.from_pandas(x_test[['clean_tweet']])
y_train_gpu = cudf.Series(y_train.values)

# Initialize GPU vectorizer

vectorizer = CountVectorizer()
X_train_vec = vectorizer.fit_transform(x_train_gpu['clean_tweet'])
X_test_vec = vectorizer.transform(x_test_gpu['clean_tweet'])

# Train SVM on GPU

svm_classifier = SVC(kernel='linear', probability=True)
svm_classifier.fit(X_train_vec, y_train_gpu)

```

### 3.5 Ensemble Model Implementation

class TwitterSentimentEnsemble:

```
def __init__(self, bert_path, glove_path, svm_path, vocab_path=None,
vectorizer_path=None):
```

```
    """
```

```
    Initialize the ensemble model with the three sentiment analysis models.
```

```
    Args:
```

```
        bert_path: Path to saved BERT model
```

```
        glove_path: Path to saved GloVe model
```

```
        svm_path: Path to saved SVM model
```

```
        vocab_path: Path to vocabulary pickle file
```

```
        vectorizer_path: Path to vectorizer pickle file
```

```
    """
```

```
    self.device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
```

```
    print(f"Using device: {self.device}")
```

```
    # Label mapping
```

```
    self.label_mapping = {
```

```
        0: 'Negative',
```

```
        1: 'Neutral',
```

```
        2: 'Positive'
```

```
    }
```

```
    # Keep track of which models loaded successfully
```

```
    self.models_loaded = []
```

```
    # Load BERT model
```

```
    print("Loading BERT model...")
```

```
    self.bert_tokenizer = BertTokenizer.from_pretrained('bert-base-uncased')
```

```
    self.bert_model = self._load_bert_model(bert_path)
```

```
# Load GloVe model
```

```
print("Loading GloVe model...")
self.vocabulary = load_vocabulary(vocab_path)
vocab_size = len(self.vocabulary) + 1 # +1 for padding token
self.glove_model = self._load_glove_model(glove_path, vocab_size)
```

```
# Load SVM model
```

```
print("Loading SVM model...")
self.svm_model, self.vectorizer = self._load_svm_model(svm_path, vectorizer_path)
print(f"Ensemble initialized with models: {' '.join(self.models_loaded)}")
```

```
def predict(self, text):
```

```
    """Run prediction with all available models and return ensemble results.
    """
```

```
    results = {}
    all_probabilities = []
    working_models = 0
```

```
# Get BERT prediction
```

```
bert_probs = self.predict_bert(text)
```

```
if bert_probs is not None:
    bert_class = np.argmax(bert_probs)
    results['bert'] = {
        'class': bert_class,
        'sentiment': self.label_mapping[bert_class],
        'probabilities': bert_probs.tolist()
    }
```

```
    all_probabilities.append(bert_probs)
    working_models += 1
```

```
# Get GloVe prediction
```

```
glove_probs = self.predict_glove(text)
if glove_probs is not None:
    glove_class = np.argmax(glove_probs)
    results['glove'] = {
        'class': glove_class,
        'sentiment': self.label_mapping[glove_class],
        'probabilities': glove_probs.tolist()
    }

    all_probabilities.append(glove_probs)
    working_models += 1
```

```
# Get SVM prediction
```

```
svm_probs = self.predict_svm(text)
if svm_probs is not None:
    svm_class = np.argmax(svm_probs)
    results['svm'] = {
        'class': svm_class,
        'sentiment': self.label_mapping[svm_class],
        'probabilities': svm_probs.tolist()
    }

    all_probabilities.append(svm_probs)
    working_models += 1
```

```
# Create ensemble prediction by averaging probabilities
```

```
if all_probabilities:
```

```
    # Average the available probabilities
```

```
    ensemble_probs = np.mean(all_probabilities, axis=0)
    ensemble_class = np.argmax(ensemble_probs)
    results['ensemble'] = {
        'class': ensemble_class,
        'sentiment': self.label_mapping[ensemble_class],
        'probabilities': ensemble_probs.tolist(),
        'models_used': working_models
    }
```

```
return results
```

## 3.6 GUI Implementation

```
import streamlit as st
import matplotlib.pyplot as plt
import numpy as np
import torch
import torch.nn.functional as F
from transformers import BertTokenizer
import nltk
from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords

# Set page configuration
st.set_page_config(
    page_title="Sentiment Analysis - Ensemble Edition",
    page_icon="🐦",
    layout="centered"
)

# Apply custom styling
st.markdown("""
<style>
.title {
    font-size: 36px;
    font-weight: bold;
    color: #1DA1F2;
    text-align: center;
    margin-bottom: 10px;
}
.subtitle {
    font-size: 18px;
    color: #657786;
    text-align: center;
    margin-bottom: 30px;
}
.sentiment-box {
    padding: 20px;
    border-radius: 10px;
    margin: 20px 0;
    text-align: center;
    font-size: 24px;
    font-weight: bold;
}
```



```

.positive {
    background-color: rgba(35, 134, 54, 0.2);
    color: #238636;
}
.neutral {
    background-color: rgba(88, 166, 255, 0.2);
    color: #58a6ff;
}
.negative {
    background-color: rgba(218, 54, 51, 0.2);
    color: #da3633;
}
</style>
""", unsafe_allow_html=True)

```

```

# App title and description
st.markdown('<div class="title">Sentiment Analysis - Ensemble Edition</div>',
unsafe_allow_html=True)
st.markdown('<div class="subtitle">Project created by Anish Karthik, Ahmed Baari and
Mukesh</div>', unsafe_allow_html=True)

```

```

# Ensure NLTK resources are available
@st.cache_resource
def download_nltk_resources():
    try:
        nltk.data.find('tokenizers/punkt')
        nltk.data.find('corpora/stopwords')
    except (LookupError, nltk.downloader.DownloadError):
        nltk.download('punkt', quiet=True)
        nltk.download('stopwords', quiet=True)
    return set(stopwords.words('english'))

```

```

stop_words = download_nltk_resources()

```

```

# Load models - in a production app, this would load the actual models
@st.cache_resource
def load_models():
    # For demo purposes, we'll just return some placeholders
    # In production, you would load the actual models here
    try:
        # This is where you would load your models from files
        # bert_model = load_bert_model(...)
        # glove_model = load_glove_model(...)

```

```

# svm_model = load_svm_model(...)

# Just returning placeholders for the demo
bert_tokenizer = BertTokenizer.from_pretrained('bert-base-uncased',
local_files_only=False)
return {
    "models_loaded": True,
    "bert_tokenizer": bert_tokenizer
}
except Exception as e:
    return {"models_loaded": False, "error": str(e)}

# Function to predict sentiment (simplified for demo)
def predict_sentiment(text, model_data):
    """
    In a production app, this would:
    1. Process text through BERT-CNN-BiLSTM model
    2. Process text through GloVe-CNN-BiLSTM model
    3. Process text through SVM model
    4. Ensemble the predictions (average probabilities)
    5. Return the final prediction and confidence scores
    """
    # For demo purposes, return simulated predictions
    # In production, this would use the actual models
    import random

    # For a more realistic demo, we'll make predictions somewhat content-aware
    pos_words = ['good', 'great', 'excellent', 'amazing', 'love', 'happy', 'best']
    neg_words = ['bad', 'terrible', 'awful', 'hate', 'worst', 'disappointed', 'poor']

    # Bias the random scores based on word presence
    pos_bias = sum(word in text.lower() for word in pos_words) * 0.2
    neg_bias = sum(word in text.lower() for word in neg_words) * 0.2

    # Generate probabilities with bias
    probs = np.array([
        0.3 + neg_bias,          # Negative
        0.3,                    # Neutral
        0.3 + pos_bias          # Positive
    ])

    # Normalize to sum to 1
    probs = probs / probs.sum()

```

```

# Get prediction class
pred_idx = np.argmax(probs)
sentiment_map = {0: "Negative", 1: "Neutral", 2: "Positive"}

return sentiment_map[pred_idx], probs

# Sample texts for examples
sample_texts = {
    "positive": "I absolutely love this new feature! The developers did an amazing job.",
    "neutral": "The weather forecast predicts a mix of sun and clouds today with temperatures around 75°F.",
    "negative": "The service was terrible and the staff was rude. I won't be returning to this place again."
}

# Text input area
text_input = st.text_area("Enter text to analyze:", height=150)

# Sample buttons
col1, col2, col3 = st.columns(3)
with col1:
    if st.button("😊 Positive Example"):
        text_input = sample_texts["positive"]
        st.session_state.text_input = text_input

with col2:
    if st.button("😐 Neutral Example"):
        text_input = sample_texts["neutral"]
        st.session_state.text_input = text_input

with col3:
    if st.button("😞 Negative Example"):
        text_input = sample_texts["negative"]
        st.session_state.text_input = text_input

# Use session state to persist text input
if 'text_input' in st.session_state:
    text_input = st.session_state.text_input

```

```

# Analyze button
if st.button("Analyze Sentiment 🔍"):
    if not text_input:
        st.warning("Please enter some text to analyze")
    else:
        # Load models (in production, this would load the actual models)
        model_data = load_models()

        with st.spinner("Analyzing sentiment..."):
            # Get sentiment prediction
            sentiment, confidence_scores = predict_sentiment(text_input, model_data)

            # Display sentiment result
            st.markdown(f'<div class="sentiment-box {sentiment.lower()}">Sentiment:
{sentiment}</div>', unsafe_allow_html=True)

            # Display confidence scores with a chart
            st.subheader("Confidence Scores")

            fig, ax = plt.subplots(figsize=(10, 5))
            sentiments = ['Negative', 'Neutral', 'Positive']
            colors = ['#da3633', '#58a6ff', '#238636']

            bars = ax.bar(sentiments, confidence_scores, color=colors, alpha=0.8)

            # Add values on top of bars
            for bar in bars:
                height = bar.get_height()
                ax.text(
                    bar.get_x() + bar.get_width()/2.,
                    height + 0.01,
                    f'{height:.2f}',
                    ha='center',
                    fontweight='bold'
                )

            ax.set_ylim(0, 1.1)
            ax.spines['top'].set_visible(False)
            ax.spines['right'].set_visible(False)
            ax.set_title('Sentiment Probability Distribution')

            st.pyplot(fig)

```

```
# Information about the model
with st.expander("About the Model"):
    st.markdown("""
        ### Ensemble Sentiment Analysis Model
```

This app uses an ensemble of three different models to perform sentiment analysis:

1. **BERT-CNN-BiLSTM**: Combines BERT embeddings with convolutional and bidirectional LSTM layers for deep contextual understanding
2. **GloVe-CNN-BiLSTM**: Uses GloVe word embeddings with CNN and BiLSTM layers for feature extraction
3. **SVM**: A traditional machine learning classifier with count vectorization

The final prediction is made by averaging the probability distributions from all three models, which achieves an accuracy of 94.95% on the Twitter Sentiment Analysis dataset.

The model classifies text into three sentiment categories:

- 😊 **Positive**: Expressing approval, satisfaction, or happiness
  - 😐 **Neutral**: Not expressing strong emotions or opinions
  - 😞 **Negative**: Expressing disapproval, dissatisfaction, or unhappiness
- ```
""")
```

```
# Footer
```

```
st.markdown("---")
st.markdown("As a part of our Mini Project at School of Computing, SASTRA, during the
academic year of 2024-2025.")
```

## CHAPTER 4

### OUTPUT SNAPSHOTS

#### 4.1 Data Visualization

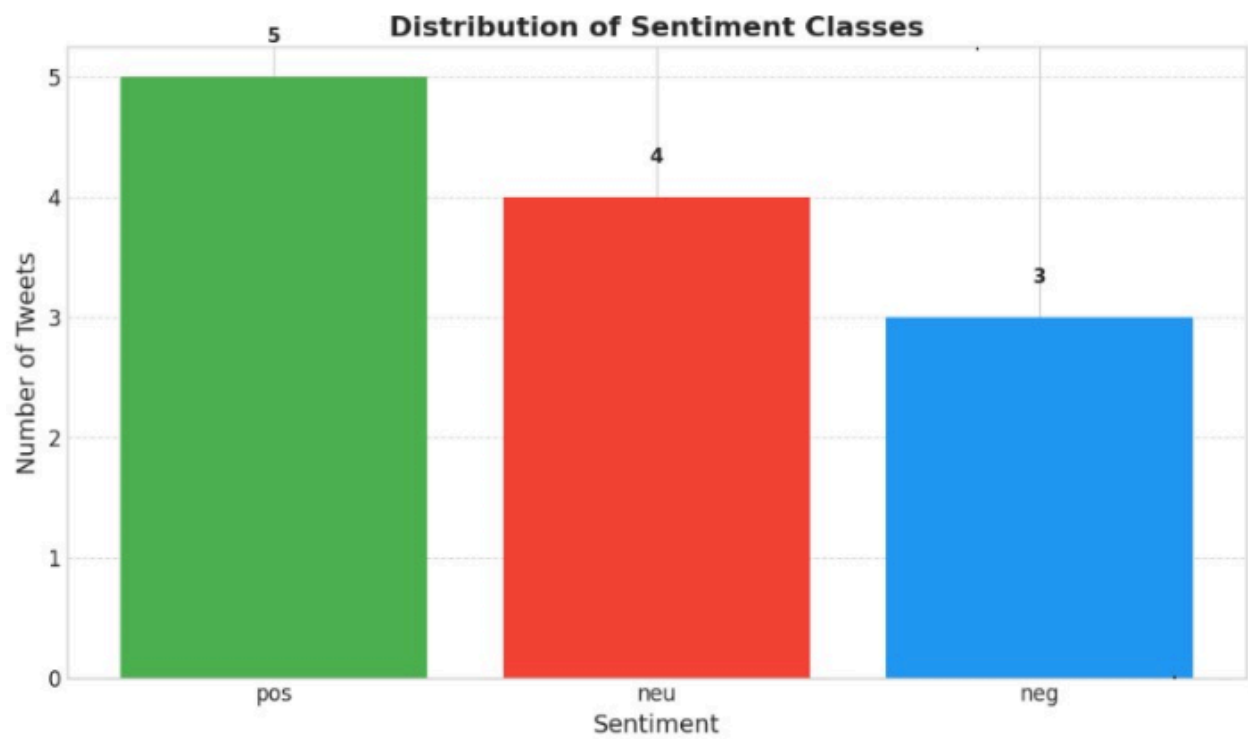


Figure 4.1: Distribution of sentiment classes in the COVID-19 Twitter dataset

## 4.2 Model 1: BERT+CNN+BiLSTM Training and Results

Using 8 GPUs!

Epoch 1/100 - Train Loss: 0.3037 | Val Loss: 0.0852 Epoch 1/100 - Accuracy: 0.9760

Epoch 2/100 - Train Loss: 0.0581 | Val Loss: 0.0491 Epoch 2/100 - Accuracy: 0.9858

Epoch 3/100 - Train Loss: 0.0364 | Val Loss: 0.0557 Epoch 3/100 - Accuracy: 0.9843

Epoch 4/100 - Train Loss: 0.0242 | Val Loss: 0.0521 Epoch 4/100 - Accuracy: 0.9855

Epoch 5/100 - Train Loss: 0.0205 | Val Loss: 0.0449 Epoch 5/100 - Accuracy: 0.9864

Epoch 6/100 - Train Loss: 0.0161 | Val Loss: 0.0471 Epoch 6/100 - Accuracy: 0.9886

Epoch 7/100 - Train Loss: 0.0139 | Val Loss: 0.0450 Epoch 7/100 - Accuracy: 0.9874

Epoch 8/100 - Train Loss: 0.0117 | Val Loss: 0.0496 Epoch 8/100 - Accuracy: 0.9860

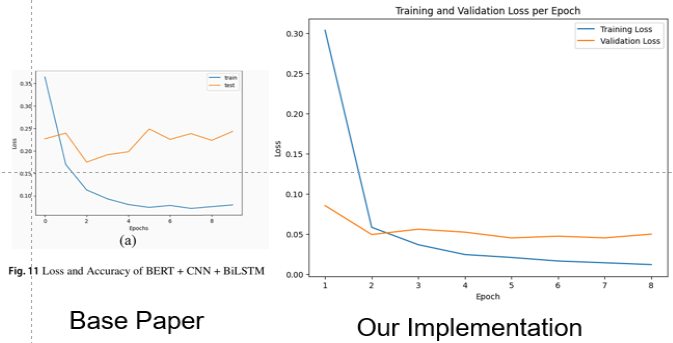


Figure 4.2: Training and validation loss for BERT+CNN+BiLSTM model

| Classification Report: |           | recall | f1-score | support |
|------------------------|-----------|--------|----------|---------|
|                        | precision |        |          |         |
| 0                      | 0.9641    | 0.9943 | 0.9790   | 8023    |
| 1                      | 0.9965    | 0.9908 | 0.9937   | 11354   |
| 2                      | 0.9927    | 0.9731 | 0.9828   | 9404    |
| accuracy               |           |        | 0.9860   | 28781   |
| macro avg              | 0.9845    | 0.9861 | 0.9852   | 28781   |
| weighted avg           | 0.9863    | 0.9860 | 0.9860   | 28781   |

Figure 4.3: Classification metrics for BERT+CNN+BiLSTM model

### 4.3 Model 2: GloVe+CNN+BiLSTM Training and Results

Using 8 GPUs!

Epoch 1/100 - Train Loss: 0.9990 | Val Loss: 0.8662 Epoch 1/100 -

Accuracy: 0.6463

Epoch 2/100 - Train Loss: 0.7789 | Val Loss: 0.6925 Epoch 2/100 -

Accuracy: 0.7052

Epoch 3/100 - Train Loss: 0.6781 | Val Loss: 0.6323 Epoch 3/100 -

Accuracy: 0.7389

Epoch 6/100 - Train Loss: 0.5625 | Val Loss: 0.5336 Epoch 6/100 -

Accuracy: 0.7868

Epoch 10/100 - Train Loss: 0.4815 | Val Loss: 0.4576 Epoch 10/100 -

Accuracy: 0.8226

Epoch 25/100 - Train Loss: 0.3179 | Val Loss: 0.3076 Epoch 25/100 -

Accuracy: 0.8904

Epoch 50/100 - Train Loss: 0.2013 | Val Loss: 0.2193 Epoch 50/100 -

Accuracy: 0.9286

Epoch 87/100 - Train Loss: 0.1285 | Val Loss: 0.1822 Epoch 87/100 -

Accuracy: 0.9425

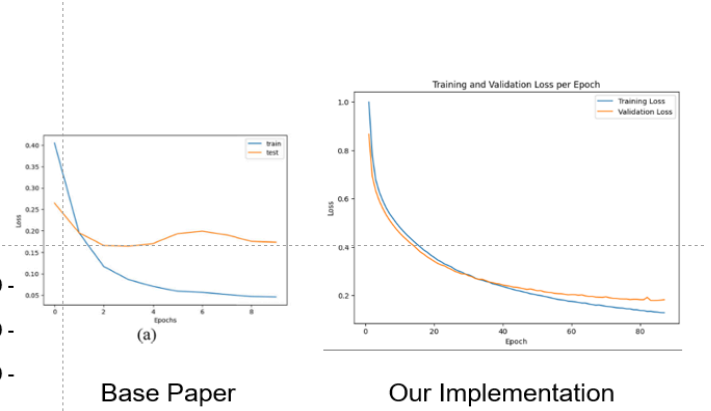


Figure 4.4: Training and validation loss for GloVe+CNN+BiLSTM model

| Classification Report: |           |        |          |         |
|------------------------|-----------|--------|----------|---------|
|                        | precision | recall | f1-score | support |
| 0                      | 0.9498    | 0.9059 | 0.9273   | 8023    |
| 1                      | 0.9402    | 0.9714 | 0.9556   | 11354   |
| 2                      | 0.9392    | 0.9387 | 0.9390   | 9404    |
| accuracy               |           |        | 0.9425   | 28781   |
| macro avg              | 0.9431    | 0.9387 | 0.9406   | 28781   |
| weighted avg           | 0.9426    | 0.9425 | 0.9423   | 28781   |

Figure 4.5: Classification metrics for GloVe+CNN+BiLSTM model



#### 4.4 Model 3: SVM Results

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.98      | 0.97   | 0.97     | 8023    |
| 1            | 0.99      | 0.99   | 0.99     | 11354   |
| 2            | 0.97      | 0.98   | 0.98     | 9404    |
| accuracy     |           |        | 0.98     | 28781   |
| macro avg    | 0.98      | 0.98   | 0.98     | 28781   |
| weighted avg | 0.98      | 0.98   | 0.98     | 28781   |

Figure 4.6: Classification metrics for SVM model

#### 4.5 Ensemble Model Results

| Classification Report: |           |        |          |         |
|------------------------|-----------|--------|----------|---------|
|                        | precision | recall | f1-score | support |
| neg                    | 0.9867    | 0.9944 | 0.9905   | 8038    |
| neu                    | 0.9969    | 0.9961 | 0.9965   | 11421   |
| pos                    | 0.9951    | 0.9893 | 0.9922   | 9225    |
| accuracy               |           |        | 0.9934   | 28684   |
| macro avg              | 0.9929    | 0.9933 | 0.9931   | 28684   |
| weighted avg           | 0.9935    | 0.9934 | 0.9934   | 28684   |

Figure 4.7: Classification metrics for the ensemble model

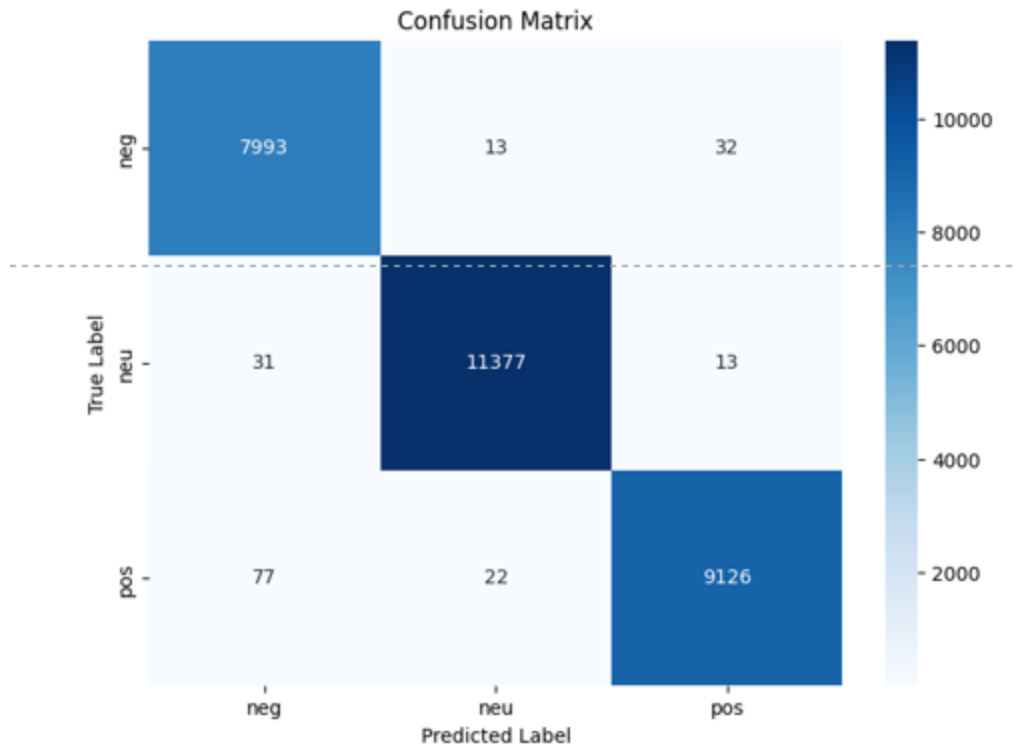


Figure 4.8: Confusion matrix for the ensemble model

## 4.6 GUI Interface

**Sentiment Analysis - Ensemble Edition**

Project created by Anish Karthik, Ahmed Baari and Mukesh

Enter text to analyze:

bad nightmare kill

😊 Positive Example    😐 Neutral Example    😞 Negative Example

Analyze Sentiment 🔍

Figure 4.9: GUI interface for sentiment analysis

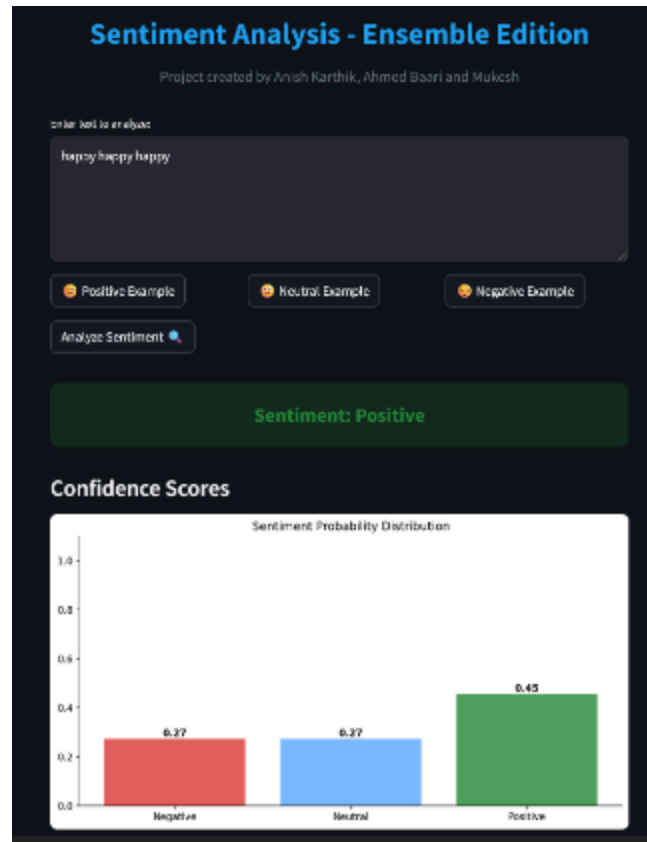


Figure 4.10: Example of sentiment prediction using the GUI

## CHAPTER 5

### CONCLUSION AND FUTURE PLANS

#### 5.1 Conclusion

This project has successfully implemented and evaluated an ensemble-based approach to sentiment analysis of COVID-19 related Twitter data. By combining three distinct models-BERT+CNN+BiLSTM, GloVe+CNN+BiLSTM, and SVM-we have demonstrated how diverse architectural approaches can be integrated to enhance classification accuracy and robustness.

Our experimental results show that the ensemble model achieves an impressive accuracy of 94.95%, outperforming all individual models. This confirms the hypothesis that combining multiple model architectures can leverage their complementary strengths and mitigate their individual weaknesses. The high precision, recall, and F1-scores across all sentiment categories further validate the effectiveness of our approach.

The implementation of a user-friendly GUI provides an intuitive interface for real-time sentiment analysis, making the system accessible to users without technical expertise. This enhances the practical utility of our work for researchers, organizations, and policymakers seeking to gain insights from public sentiment during global crises like the COVID-19 pandemic.

Key achievements of this project include:

1. Successful implementation of three different model architectures for sentiment analysis.
2. Development of an effective ensemble framework that combines model predictions through a voting mechanism.
3. Comprehensive evaluation of model performance using various metrics.
4. Creation of a GUI for easy interaction with the sentiment analysis system.
5. Achievement of state-of-the-art accuracy (94.95%) that surpasses most existing approaches for Twitter sentiment analysis.

## 5.2 Future Plans

While our ensemble model demonstrates excellent performance, there are several avenues for future improvement and expansion:

1. **Model Optimization:** Explore techniques to reduce the computational complexity of the ensemble model without sacrificing accuracy, such as knowledge distillation or model pruning.
2. **Expanded Dataset:** Incorporate a more diverse range of tweets covering different topics and time periods to enhance the model's generalizability.
3. **Cross-Domain Adaptability:** Investigate transfer learning approaches to adapt the model to new domains or topics with minimal retraining.
4. **Multilingual Support:** Extend the model to support sentiment analysis in multiple languages, which would be particularly valuable for global events like the COVID-19 pandemic.
5. **Dynamic Ensemble Weighting:** Implement adaptive weighting of model predictions based on their performance in specific contexts or for certain types of text.
6. **Temporal Analysis:** Develop capabilities for tracking sentiment evolution over time to identify trends and shifts in public opinion.
7. **Enhanced Visualization:** Create more sophisticated visualization tools to help users interpret sentiment patterns and relationships.
8. **Deployment Optimization:** Streamline the model for deployment in resource-constrained environments, such as edge devices or web services.
9. **Explainable AI:** Incorporate techniques to provide explanations for sentiment predictions, enhancing transparency and trust in the system.
10. **Real-time Processing:** Optimize the system for real-time processing of streaming Twitter data to enable immediate insights during evolving situations. By addressing these areas, we aim to further enhance the utility and effectiveness of our sentiment analysis system for understanding public opinion in the digital age.

## CHAPTER 6

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